Can XR Trust Its Nose? A Preliminary Study of Sensing Integrity for Olfactory Interfaces

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Abstract

Olfactory extended reality (XR) systems increasingly combine scent output with electronic noses (e-noses) that sense odor context. Yet an e-nose reading is only a situated proxy for what a user smells: odor transport can be delayed, uneven, persistent, and shifted by airflow, humidity, sensor history, and mixtures. This paper studies **sensing integrity** for olfactory XR, asking whether a current odor reading is valid for the user's exposure, virtual event, and next system action. We ran a preliminary study with 30 participants across six scenarios: safety training, cooking, collaborative laboratory work, medical exposure, museum visiting, and relaxation. Each scenario paired a controlled real-room odor manipulation with a matched VR task and questionnaire. The results identify recurring failures–latency, residual scent, spatial/airflow mismatch, humidity shift, occlusion, mixtures, and action-policy risk–which we organize into a taxonomy for future olfactory-interface design and evaluation.

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CCS Concepts

• **Human-centered computing** → **Ubiquitous and mobile computing systems and tools**; **Empirical studies in interaction design**.

Keywords

olfactory interfaces, electronic nose, virtual reality, augmented reality, sensing integrity

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1 Introduction

Extended reality (XR) systems are moving beyond visual and auditory immersion toward multisensory interaction, including haptics, temperature, airflow, and smell. Olfactory cues can strengthen place, memory, and realism in virtual environments, and recent systems have explored scent delivery for narrative virtual reality (VR), affective interfaces, and miniaturized wireless smell displays [2, 29, 33, 39]. At the same time, emerging XR agents and context-aware applications [11] increasingly need to sense what is happening around the user while commanding scent output. This

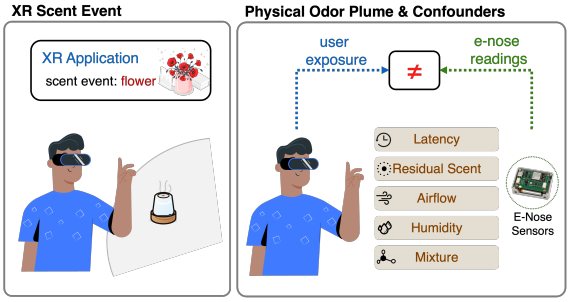


Figure 1: Sensing integrity asks when user exposure and e-nose readings diverge enough to change an XR action.

paper focuses on olfactory XR systems that both deliver scents to the user and use a compact electronic nose (e-nose) as an input or context sensor. The core problem is alignment: the application must know whether the odor sampled by a headset-mounted, room, prop, or robot-mounted e-nose represents what the user is smelling when the XR policy acts.

Low-cost volatile organic compound (VOC) and gas sensors can be small enough for headsets, rooms, props, or mobile robots, but interactive XR turns a direct odor-labeling interface into a policy risk because chemical sensing is slow, local, and stateful. In safety training, a blocked inlet or off-axis plume can falsely mark a path as safe. In cooking, residual coffee or vinegar can belong to the previous recipe step after the virtual scene has advanced. In a collaborative lab, a fixed sensor can broadcast a shared "safe" status while a nearby user stands in a different plume. In medical exposure, slow recovery can change the intended exposure dose. In a museum visit or relaxation scene, mixed scents, airflow, or humidity can make the system overstate authenticity or warn at the wrong moment. In these cases, a classifier can be confidently wrong in exactly the moments when an XR system needs to decide whether to continue, pause, purge, warn, or ask for another sample. Figure 1 illustrates this mismatch between user exposure, e-nose readings, and XR policy choices.

We use **sensing integrity** in this paper to denote the action-conditioned trustworthiness of the current olfactory belief. Related integrity concepts appear in safety-critical active sensing and smart-home functional-integrity verification, where a sensing stream must remain reliable enough for a downstream safety or security action [5, 45]. Traditional sensing robustness and reliability ask whether a sensor stream remains calibrated, stable, and fault-free under noise, drift, environmental variation, or data-integrity attacks [12, 30, 32, 41]. Our use places this integrity framing in olfactory XR: the system must estimate whether the current reading is valid for a specific user, virtual event, physical sampling path, and policy action. A calibrated sensor can still have low integrity when it is recovering from a previous scent, sampling behind a fan, or observing a local plume that misses the user. A noisy reading can still be sufficient for a low-risk relaxation cue if the policy only needs to mark uncertainty. The key question extends from "what odor is present?" to "is this reading valid for the next XR action?" This distinction matters in safety training, medical or exposure simulation, museum visits, collaborative augmented reality (AR) laboratories, and artificial intelligence (AI) agents that may store olfactory context as memory.

This paper studies sensing integrity as a user-facing problem before proposing a full sensing system. We design a pilot study with real-room odor manipulations, matched VR scenarios, and post-scenario questionnaires across six XR contexts: safety training, cooking, collaborative laboratory work, medical exposure, museum visiting, and relaxation. We use the quantitative ratings and open responses to derive an XR-specific failure taxonomy.

Our contributions are:

- We operationalize sensing integrity for olfactory XR and organize common failures into temporal, spatial, environmental, sensor-state, semantic, and action-policy categories.
- We design a controlled pilot study with real-room odor manipulations, matched VR scenarios, and questionnaires to ground sensing-integrity concerns in user-facing olfactory-XR situations.
- We derive an XR-specific failure taxonomy and design implications that future olfactory interfaces can use to test whether an e-nose reading is valid for rendering, delaying, purging, resampling, warning, or stopping.

2 Related Work

Olfactory Interfaces and Extended Reality. Prior work establishes smell as an interaction modality for affect, information, memory, agency, and immersion [18, 33, 38, 43]. Interactive systems show effects on mood, cognition, information transfer, memory reactivation, agency, writing, and multisensory virtual reality (VR) [2–4, 10, 16, 39]. Toolkits and devices make scent easier to prototype or deliver, including low-fidelity authoring, task-oriented development, fabricated release structures, localized heating, on-face or soft wireless displays, gesture-triggered release, breath-aware interaction, and room-scale mixed reality [7, 13, 14, 22, 24, 25, 29, 48–50, 52]. These systems show why scent matters for extended reality (XR); our study examines when a sensed olfactory belief is trustworthy enough for a user-facing action.

Odor Sensing and Integrity. Compact chemical-sensing systems classify odors, monitor activities, and measure chemical context. Examples include commodity-gas odor classification [17], kitchen odor monitoring [19], infrared and terahertz mechanisms [27, 31], city-scale and wearable air sensing [9, 28, 42, 44, 47], and small broad-response volatile organic compound (VOC) sensors such as BME688 and SGP40 [6, 40]. Reliability work further shows that chemical readings require calibration, context, and validation [8, 12, 15, 21, 23, 30, 32, 34, 36, 41]. Related integrity language appears in active roadway sensing and smart-home functional verification [5, 45], while work on electronic noses (e-noses) and plume sensing explains cross-sensitivity, recovery, drift, interference, and local plume transport [20, 26, 37, 46]. These papers ask whether a sensor stream is calibrated and reliable for labels, maps, alarms, or offline decisions. Olfactory XR adds a user- and task-grounded question: whether the current e-nose belief belongs to the user's virtual event, physical exposure, and next action. Our pilot derives the taxonomy from scent manipulations, VR scenarios, and questionnaires.

3 Pilot Study

We conducted a pilot study to ground sensing integrity in user-facing olfactory XR situations. The goal is to understand sensing integrity by crossing controlled real-room odor manipulations with matched VR tasks and post-scenario questionnaires. The six scenarios vary temporal state, spatial sampling, environmental condition, semantic mixture, and action risk, allowing us to observe how physical scent behavior, user experience, and policy preference interact when the electronic nose (e-nose) reading and the experienced scent field diverge.

Participants. We recruited 30 participants between 19 and 52 years old. The group included 15 self-identified men and 15 self-identified women, with occupations spanning students, professors, engineers, accountants, physicians, and nurses. Each participant completed six scenarios. Before the first scenario, an experimenter gave a 10 min briefing on the VR headset, safety constraints, and study goal. Before each scenario, the experimenter gave a 5 min briefing on the task goal, rating scales, participant actions, and options to render, delay, purge, resample, or warn.

Procedure. Each scenario lasted 15 min. Participants first experienced a controlled real-room odor condition, then entered a matched VR scene in which an XR application displayed one policy selected from the simulated e-nose state: continue the task, delay the next step, purge, resample, warn, or stop. After each scenario, the participant moved to an adjacent room for 30 min. During this period, the participant completed the scenario questionnaire and rested. Meanwhile, the experiment room was reset: for the first 15 min, windows were opened and fans exchanged the air to reduce carryover; for the next 15 min, windows were closed and the air conditioner, humidifier, fan, and sensor logger restored the target temperature, humidity, and airflow; scenario order was counterbalanced.

Environment. The study was conducted in a meeting room. We used safe volatile organic compound (VOC)-rich everyday odor sources such as coffee, vinegar, citrus peel or citrus oil, tea extract, and alcohol wipes, plus neutral airflow and humidity perturbations. The sensing node used a compact BME688 metal-oxide (MOX) module with temperature and humidity logging [6]. The logger recorded gas/VOC response, temperature, humidity, fan state, source-sensor distance, and event markers for each scenario.

Source labels identify the trial conditions; the BME688 response is used to examine timing, recovery, and sampling alignment rather than to establish a validated classifier for each everyday scent.

VR scenes. The six matched VR tasks used scenario-specific scenes to make each integrity failure actionable for participants: safety training, cooking, collaborative lab, medical exposure, museum visit, and relaxation (Figure 2)[1].

Questionnaire and analysis. The questionnaire combined 7-point Likert items, action-choice questions, cue attribution, and open comments. Quantitative items asked about scent-event synchronization, perceived integrity concern, trust decrease, and preferred policy action. Open comments were coded into seven failure cues: *latency*, *residual carryover*, *spatial mismatch*, *temperature or humidity shift*, *occlusion*, *out-of-distribution (OOD) odor*, meaning an odor or mixture outside the expected scenario set, and *action-policy risk*.

[1]The VR scenes were developed in Unity; the 3D models used in the scenes were downloaded from open-source assets on the Unity Asset Store.

Section 4 uses the quantitative trends and open-response cues to construct the failure taxonomy. Appendix A lists the questionnaire items.

3.1 Safety Training

Real environment. Participants stood in the meeting room when a vinegar or alcohol-wipe source was introduced near one side of the room. A fan changed plume direction, and a simple cover partially blocked the sensor inlet during part of the trial. The participant stood alternately closer to or farther from the source than the sensor, making the local reading diverge from user exposure.

VR environment. The matched VR task placed the participant in an industrial safety-training scene. A virtual leak, warning light, and blocked doorway required the system to decide whether the user could proceed. The e-nose decision was shown as an XR policy action: continue the task, delay the next step, request resampling, warn the user, or stop the task.

Design goal. This scenario targets high-risk false-safe behavior. It tests whether users consider a confident, spatially invalid or occluded sensor reading more serious when the virtual consequence is entering a hazardous area. It also elicits the relative preference for warning, stopping, and delaying the scent event.

Survey. The key quantitative questions asked whether the system's safety judgment was trustworthy, whether the participant preferred a conservative action, and how severe the failure would be in real training. Open comments were used to identify spatial mismatch, occlusion, and action-risk language.

3.2 Cooking

Real environment. Participants experienced a sequence of coffee, vinegar, citrus peel or citrus oil, and alcohol-wipe sources with short gaps between events. In some trials the next source was introduced before the previous odor fully cleared, creating residual carryover. The sensor and participant were kept at a fixed distance to make timing and recovery the main manipulation.

VR environment. The VR task simulated a cooking lesson. The participant followed recipe steps such as roasting coffee, adding vinegar, or checking whether a food item was fresh. The virtual scene sometimes advanced faster than the physical odor field.

Design goal. This scenario studies latency mismatch and residual carryover. In cooking, odor is meaningful and usually low-to-medium risk, making it a good test of whether users prefer delay or purge over a hard stop.

Survey. Participants rated whether the smell matched the current visual step, whether the previous scent contaminated the next step, and whether a scene delay would be acceptable. Open comments were coded for latency, residual carryover, and semantic confusion.

3.3 Collaborative Lab

Real environment. Two positions were marked in the meeting room: a user position and a sensor position. The odor source and fan were arranged so that one position received a stronger plume than the other. In part of the scenario, the participant moved between positions while the sensor stayed fixed.

VR environment. The VR task represented a collaborative laboratory in which the user and a virtual teammate shared air-quality or



(a) Real-room setup (b) Safety (c) Cooking (d) Lab (e) Medical (f) Museum (g) Relaxation

Figure 2: Illustrative physical setup and six olfactory XR scenarios.

chemical-context information. The system had to decide whether to broadcast a shared "safe" or "unsafe" status.

Design goal. This scenario targets spatial mismatch and multi-user consequences. A reading that is locally valid for the sensor may be invalid for a nearby user or teammate. The study asks whether participants want the system to resample, request another viewpoint, or display uncertainty before sharing context.

Survey. Participants rated whether they trusted a single fixed sensor, whether they preferred multi-point confirmation, and whether sharing an uncertain odor state with collaborators was acceptable. Open comments captured spatial mismatch, airflow, and policy risk.

3.4 Medical Exposure

Real environment. Participants experienced low-intensity, safe odor exposure with explicit purge periods. The room was reset between trials; selected trials deliberately started before a complete return to baseline to mimic residual exposure. The temperature and humidity targets were held stable so that recovery state and policy risk dominated the scenario.

VR environment. The VR task represented a medical or exposure-therapy simulation in which the system moved through staged exposure levels. The simulated e-nose determined whether the next stage should begin, pause, or be purged and repeated.

Design goal. This scenario studies the action-conditioned nature of integrity. The same residual reading that may be tolerable in cooking can be unacceptable in a medical or therapeutic context because residual exposure changes the intended dose and perceived control.

Survey. Participants rated acceptable false alarms and missed exposure cues, whether the system should wait for a stricter recovery threshold, and whether warning was sufficient or the task should stop. Open comments were coded for residual carryover, latency, and policy risk.

3.5 Museum Visit

Real environment. Participants experienced pretested VOC-rich scent jars such as tea extract, citrus peel or citrus oil, coffee, and mild wood-like essential-oil blends in a room with a low background odor. Some trials mixed two sources to create an unfamiliar scent that resembled the expected scene odor while diverging from it.

VR environment. The VR task presented a museum scene in which scent was used to enhance a historical room, market, or artifact. The system could render the scent, mark uncertainty, delay, or resample.

Design goal. This scenario targets semantic mismatch and OOD mixtures in an immersion-oriented setting. The study asks whether users value strict correctness or whether they accept approximate scents when the consequence is immersion rather than safety.

Survey. Participants rated immersion, perceived authenticity, and whether an unfamiliar or mixed scent should be treated as an error. Open comments were coded for OOD odor, residual background odor, and low-risk policy preferences.

3.6 Relaxation

Real environment. Participants sat in the room while a mild pleasant scent was introduced at low intensity. The main perturbation was a controlled temperature or humidity change from the air conditioner or humidifier while the odor source remained fixed. This created sensor variation that could look like an odor event under a stable intended scent scene.

VR environment. The VR task was a relaxation or meditation scene, such as a forest or spa environment. The system decided whether to keep rendering, mark uncertainty, or softly warn that the scent state was unreliable.

Design goal. This scenario targets environmental shifts and low-risk policy behavior. It tests whether users prefer unobtrusive uncertainty handling when the task is ambient and low stakes.

Survey. Participants rated whether the scent change was noticeable, whether an uncertainty label would be distracting, and whether the system should continue rendering under low integrity. Open comments were coded for temperature or humidity shift, perceived annoyance, and tolerance for uncertain sensing.

4 Findings and Taxonomy

The pilot connects physical odor behavior to user trust and action preference: each scenario paired a real-room manipulation with an XR consequence, then asked participants whether the system should render, delay, purge, resample, warn, or stop. This section first analyzes the questionnaire trends and then abstracts the recurring failures into a taxonomy.

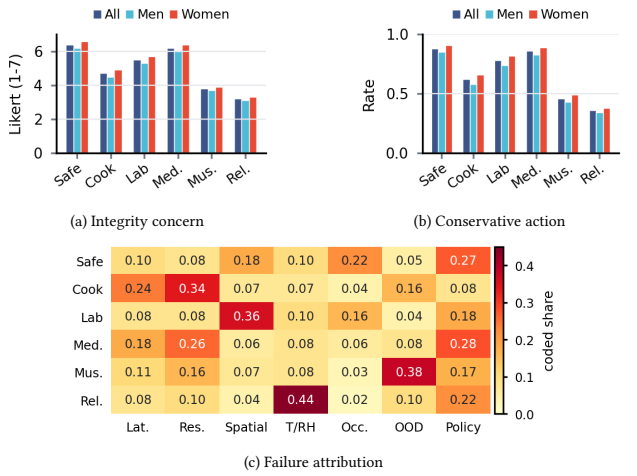


Figure 3: Questionnaire trends across six scenarios. OOD: out-of-distribution; T/RH: temperature/relative humidity.

Table 1: Sensing-integrity failure taxonomy.

Failure	Observable cue	Typical XR response
Latency mismatch	Event-to-rise delay; response outside scene window	Delay, resample, or hold next step
Residual carryover	Baseline not recovered; previous-source tail	Purge, wait, or mark uncertain
Spatial/airflow mismatch	Source-sensor-user offset; fan disagreement	Resample, warn, or localize status
Environmental cross-shift	Response co-varies with temperature, humidity, or airflow	Downweight, recalibrate, or pause
Occlusion/path failure	Flat response during an active odor event	Warn, stop, or resample
Out-of-distribution (OOD) mixture	Low margin, unusual dynamics, or high residual error	Mark uncertainty; avoid high-risk action
Action-policy mismatch	Same uncertainty has different task risk	Bind confidence to action-specific policy

4.1 Questionnaire Findings

Figure 3(a) shows that integrity concern is highest in safety training and medical exposure, followed by collaborative lab. The open responses explain why these scores are high. In safety training, participants described the blocked inlet and off-axis plume as a "false-safe" problem: the system could tell the user to enter a hazardous area while the user still smelled the leak. In medical exposure, participants focused less on odor identity and more on dose and control, noting that a residual odor could make the next stage feel involuntary. In collaborative lab, participants questioned whether one fixed sensor could speak for another person's local air. These answers imply three different integrity requirements: the reading must be timely enough for the event, spatially representative of the user, and strict enough for the consequence of the next action.

Moderate and low scores still contributed distinct information. Cooking received lower concern than safety or medical exposure, yet comments repeatedly described the smell as belonging to the "previous step" or arriving after the visual recipe had advanced. This scenario therefore exposes temporal ownership: the same sensor value may be chemically real but attached to the wrong virtual event. Museum visit had lower conservative-action preference because several participants accepted approximate scents when the goal was ambience, but the open responses still flagged unfamiliar mixtures as authenticity errors. This scenario motivates semantic integrity rather than safety integrity. Relaxation had the lowest concern and action preference, but it uniquely surfaced temperature/humidity shifts: participants tolerated continued rendering if the system communicated uncertainty. This scenario prevents the taxonomy from treating every integrity problem as a stop condition.

Figure 3(b) separates concern from action: safety and medical tasks favor conservative actions, cooking favors delay or purge, collaborative lab favors resampling, and museum and relaxation favor uncertainty marking. These differences motivate action-policy mismatch rather than treating integrity as a single global threshold.

Figure 3(c) shows that the scenarios surface different failure cues. Safety combines occlusion, spatial mismatch, and policy risk; cooking is dominated by latency and residual carryover; collaborative lab highlights spatial mismatch; medical exposure combines residual carryover with policy risk; museum visits highlight unfamiliar mixtures; and relaxation highlights temperature or humidity shifts.

Design implications. First, XR interfaces should make odor state transitions visible. Residual carryover should trigger a visual purging or recovery indicator before the virtual story advances. Second, multi-user or room-scale scenes should localize confidence rather than broadcasting one global label; a low-confidence shared status can use translucent rendering or a "synchronizing air quality" prompt until sampling matches the user region. Third, semantic uncertainty should be an interaction state: OOD mixtures can continue as marked ambience in low-risk scenes, but safety or medical tasks should block, warn, or request confirmation.

4.2 Failure Families

Construction. The taxonomy combines pilot-study cues with related sensing literature. A family captures a mismatch that can alter an XR action, has a candidate observable cue, and suggests a distinct response. The literature supplies olfactory timing and delivery constraints [1, 35, 51]; e-nose recovery, drift, and interference [26, 32, 46]; and plume, fault, and data-integrity issues [20, 30, 37, 41]. Table 1 summarizes the families and their candidate cues and responses. These cues motivate future implementation and validation.

Temporal and stateful failures. *Latency mismatch* occurs when odor reaches the user or sensor after the virtual event that requested it. The cue is an event-to-rise delay or response-window violation, and the XR action is to delay, resample, or hold the next step. *Residual carryover* occurs when the previous scent tail contaminates a new scene or keeps the sensor in recovery. The cue is incomplete baseline return, and the action is to purge, wait for recovery, or mark the scent state uncertain.

Situated sampling failures. *Spatial or airflow mismatch* occurs when a local sensor sample does not represent the user or collaborator. The cue is source-sensor-user geometry, fan state, or multi-position disagreement, and the action is to resample, warn, or display uncertainty before sharing a status. *Environmental cross-shift* occurs when temperature, humidity, or airflow changes the volatile organic compound (VOC) response without a scene change. The cue is gas-environment correlation or a large temperature/humidity derivative, and the action is to downweight, recalibrate, or pause updates. *Occlusion or sampling-path failure* occurs when an inlet is blocked while the user remains exposed. The cue is a flat response during an active odor event, and the action is to warn, stop, or resample.

Semantic and policy failures. *Out-of-distribution (OOD) mixture* occurs when a mixed or unfamiliar odor falls near a known label. The cue is low margin, unusual dynamics, or high residual error, and the action is to mark uncertainty or avoid high-risk actions. *Action-policy mismatch* occurs when the same physical uncertainty is acceptable for ambient rendering but unacceptable for safety, medical exposure, or shared status. This family separates "what odor is present" from "whether the belief is valid for the next XR action," the core requirement for sensing integrity.

5 Conclusion

Future Work. The failure taxonomy suggests that future olfactory extended reality (XR) interfaces should treat an electronic nose (e-nose) label as only one input to an action-conditioned trust decision. One potential approach is an integrity layer between the odor classifier and the XR policy: it would track recovery state, event timing, temperature and humidity shifts, airflow or placement disagreement, mixture uncertainty, and task risk, then choose whether to render, delay, purge, resample, mark uncertainty, warn, or stop. Such a layer would let low-risk ambient scenes continue under uncertainty while requiring stricter evidence for safety, medical, and collaborative actions.

Summary. This paper studies sensing integrity for olfactory interfaces through a pilot study with controlled real-room odor manipulations, matched virtual reality (VR) scenarios, and participant questionnaires. The results organize user-facing failures into temporal, spatial, environmental, sensor-state, semantic, and policy families. The taxonomy provides a compact checklist for future systems to test whether an olfactory reading is valid for the next interaction, rather than only asking which odor is present.

Acknowledgments

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A Pilot Study Questionnaire

This appendix lists the pilot-study questionnaire items for each scenario. Unless otherwise stated, Likert items used a 7-point scale, where 1 means strongly disagree or minimal severity and 7 means strongly agree or very severe.

A.1 Common Questions

(1) The scent I experienced matched the current visual or task event.
(2) The scent arrived at the right time for the VR scene.
(3) The scent from a previous event remained and affected the current event.
(4) I trusted the system's current olfactory judgment.
(5) My trust in the system decreased during this scenario.
(6) The system should continue rendering the scene smoothly.
(7) The system should delay the next scene event until the scent state is reliable.
(8) The system should purge or ventilate the scent path before continuing.
(9) The system should resample or ask for another measurement.
(10) The system should warn the user that the smell state is uncertain.
(11) The system should stop the task in this scenario.
(12) If the system made the wrong decision, how severe would the consequence be?
(13) Which cue best explains the failure? Select all that apply: late scent, residual scent, sensor and user in different places, temperature or humidity change, blocked or weak airflow, unfamiliar or mixed scent, or action risk.
(14) In one or two sentences, describe what made the system feel reliable or unreliable.

A.2 Safety Training

(1) I would trust the system's safe/unsafe judgment before entering the virtual hazard area.
(2) A false safe judgment in this scenario would be unacceptable.
(3) If the sensor is blocked or away from the plume, the system should warn or stop before continuing.
(4) I prefer the system to use a conservative threshold in safety training.
(5) Open response: What evidence would convince you that the air is safe enough to proceed?

A.3 Cooking

(1) The smell helped me understand the current recipe step.
(2) A previous smell made the current step confusing.
(3) I would accept a short scene delay if it improved scent alignment.
(4) I would prefer purge or ventilation before the next food-related odor.
(5) Open response: Which part of the cooking scene felt mistimed or contaminated?

A.4 Collaborative Lab

(1) A single fixed sensor was sufficient for judging the shared lab state.
(2) The system should account for whether the user and sensor are exposed to the same plume.
(3) I would prefer another measurement before sharing a safety status with collaborators.
(4) Uncertain odor state should be shown to collaborators.
(5) Open response: When should the system trust a user's location over a fixed sensor?

A.5 Medical Exposure

(1) The system should wait for the previous scent to clear before beginning the next exposure stage.
(2) A false exposure cue would reduce my sense of control.
(3) I prefer stricter recovery thresholds in medical or therapy scenarios.
(4) A warning is sufficient when the scent state is uncertain.
(5) Open response: What would make this exposure scenario feel safe enough to continue?

A.6 Museum Visit

(1) The scent improved immersion in the historical scene.
(2) An approximate or mixed scent was acceptable in this scenario.
(3) The system should mark uncertainty and keep the scene running when the consequence is low risk.
(4) An unfamiliar smell reduced the authenticity of the experience.
(5) Open response: When does an approximate scent become misleading?

A.7 Relaxation

(1) The scent change was noticeable during the relaxation scene.
(2) A temperature or humidity change seemed to affect the scent experience.
(3) An uncertainty warning would be distracting in this scenario.
(4) I would prefer the system to continue rendering softly under low-risk uncertainty.
(5) Open response: What is the least disruptive way for the system to communicate uncertain smell sensing?